National Oceanic Atmospheric Administration (NOAA)
National Marine Fisheries Service (NMFS)
Office of Law Enforcement (OLE)
Statement of Need (SON)

Background:

NOAA's National Marine Fisheries Service (NMFS) is responsible for the stewardship of the nation's living marine resources and their habitats. NMFS Office of Law Enforcement (OLE) is responsible for the enforcement of federal laws and international treaties relative to marine fisheries, including the inspection of seafood imports at the Nation's largest ports and investigation of seafood fraud and unlawful international trade.

The United States is the largest importer of seafood in the world by value, importing between $25-$30 billion annually. Combatting illegal trade and detection of seafood fraud is critical to prevent illegal products from entering U.S. commerce and creating unfair competition for America's native seafood industry. Identification of species is an important component in regulating the international trade in marine fisheries products. Imports of various seafood products which are not readily identifiable, such as fish filets, fish products, derivatives, etc. present challenges for enforcement to confirm species identification and legality of seafood imports.

Objectives:

This Statement of Need (SON) outlines a contract to research, develop, and field deploy rapid, genetic-based test kits to accurately identify the species of imported seafood. Seafood mislabeling and substitution represent a significant economic and public health threat, unfairly undercutting domestic U.S. producers and compromise consumer trust. By leveraging cutting-edge molecular diagnostics, OLE can establish a robust, efficient, and science-driven enforcement capacity at U.S. ports of entry, thereby protecting the integrity of the U.S. seafood market.

In FY24 the United States imported more than 762 million kilograms of shrimp, valued at more than $6 billion, via 56,019 individual shipments. OLE has no capability to rapidly test to determine whether shrimp shipments are properly labeled as either farm-raised or wild caught. In The Seafood Import Monitoring Program (SIMP) identified tuna as one of the most non-compliant imported species of relevance to the program, often lacking chain of custody documents, legal origin or vessel information. Tuna is often subject to seafood fraud, mislabeling and trafficking due to its value and limited quotas. The requirement herein for the Base Year is to develop and validate a reliable, fast, accurate assay for white shrimp (*Litopaneaeus vannamei*) and various tuna species. An additional Option Year would address another species.

General Requirements:

OLE needs to develop highly accurate, field-deployable species identification technology to support realtime decision-making aimed at intercepting mislabeled and illegal seafood products, thereby safeguarding the competitiveness of American seafood. Emerging technology includes the use of assays to perform rapid, targeted detection of species-specific DNA sequences, without the need for specialized equipment, pipetting, or refrigeration. Test results are made easily visually interpretable through a colorimetric reaction test strip, analogous to home tests for COVID19. The resulting assays provide accurate information on species identity within approximately 30 minutes without using complex laboratory equipment, enabling point-of-need verification of seafood products.

The kits will use specific technology, a specific CRISPR-based genetic identification platform developed primarily for fish and seafood species identification, to create the test strips for the kits. This contract will not use available mobile identification applications, but the COVID-like genome technology portion of the identification processes in the form of test kits. The iCatch technology uses genetic technology similar to what is desired.

Shrimp Research & Development

- Stage 1: Finalize sequence analysis for Pacific white shrimp (*Litopenaeus vannamei)* to evaluate and identify assays for identification in the context of an iCatch assay design.

- Stage 2: Develop an appropriate CRISPR-based genetic identification assay, with laminar flow detection, for Pacific white shrimp (*L. vannamei)* for application on test strips

Tuna Species Research & Development

- Stage 2: Develop an appropriate CRISPR-based genetic identification assay, with laminar flow detection, from an existing, developed genetic sequence for subsequent application on test strips, for four (4) tuna species, including: Atlantic bluefin (*Thunnus thynnus*), bigeye (*Thunnus obsesus*), yellowfin (*Thunnus alabacares*) and either Pacific bluefin or albacore sp.

Shrimp and Tuna Development to Production

- Stage 3: Field Optimization: Develop a field kit for OLE Officers to deploy system test strips in the field *and* convert development to production, initially producing 1,000 test strips and 12 Test (field) kits to include all associated equipment for deployment in the field. Intermittent delivery dates of test strips may be required, depending on shelf life.

Optional Species Development

- Finalize sequence analysis for new species agreed upon by NMFS OLE and the vendor and other similar species to evaluate and identify assays for species identification in the context of an iCatch assay design. Develop an appropriate Assay and convert development to production of field kits with intermittent delivery dates.

Deliverables

The following deliverables are requested for this contract:

1. The Contractor shall submit completed sequence analysis (raw data and alignment files), assay development (detailed notes on assay development), and laboratory validation results (detailed notes on lab validation, including complete species list and results evaluated), to the NOAA/NMFS Conservation Biology Division, Marine Forensic Laboratory, Northwest Fisheries Science Center (MFL) for review and approval. Point of Contract to be provided upon award of contract.
2. The Contractor shall provide field tests to the MFL for independent laboratory testing and validation.
3. The Contractor shall assist and/or participate in piloting the developed technology during planned port operations deploying the technology during the Field Optimization.

Confidentiality and Data Privacy

This contract may require that services contractors have access to Privacy Information. Service contractors are responsible for maintaining confidentiality of all subjects and materials and may be required to sign and adhere to a Non-disclosure Agreement (NDA).

Government Furnished Resources
The Government will not furnish any resources to the Contractor in support of this contract.

Place of Performance
Work will be conducted at vendor's facility

Period of Performance:

- <u>Shrimp and Tuna Test Kits</u> (September 18, 2026 - September 17, 2028)
    - <u>Research and Development</u>
        - Stage 1 - Completed up to nine months after award
        - Stage 2 - Completed up to six months after Stage 1 completion
    - <u>Development to Production</u>
        - Stage 3 - Completed up to nine months after Stage 2 completion
- <u>Additional Species</u> (September 18, 2028 - September 17, 2030) with completion dates similar to Shrimp above
- <u>Additional Test Strips</u> (September 18, 2028 - September 17, 2030)

Technical Point of Contact:

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